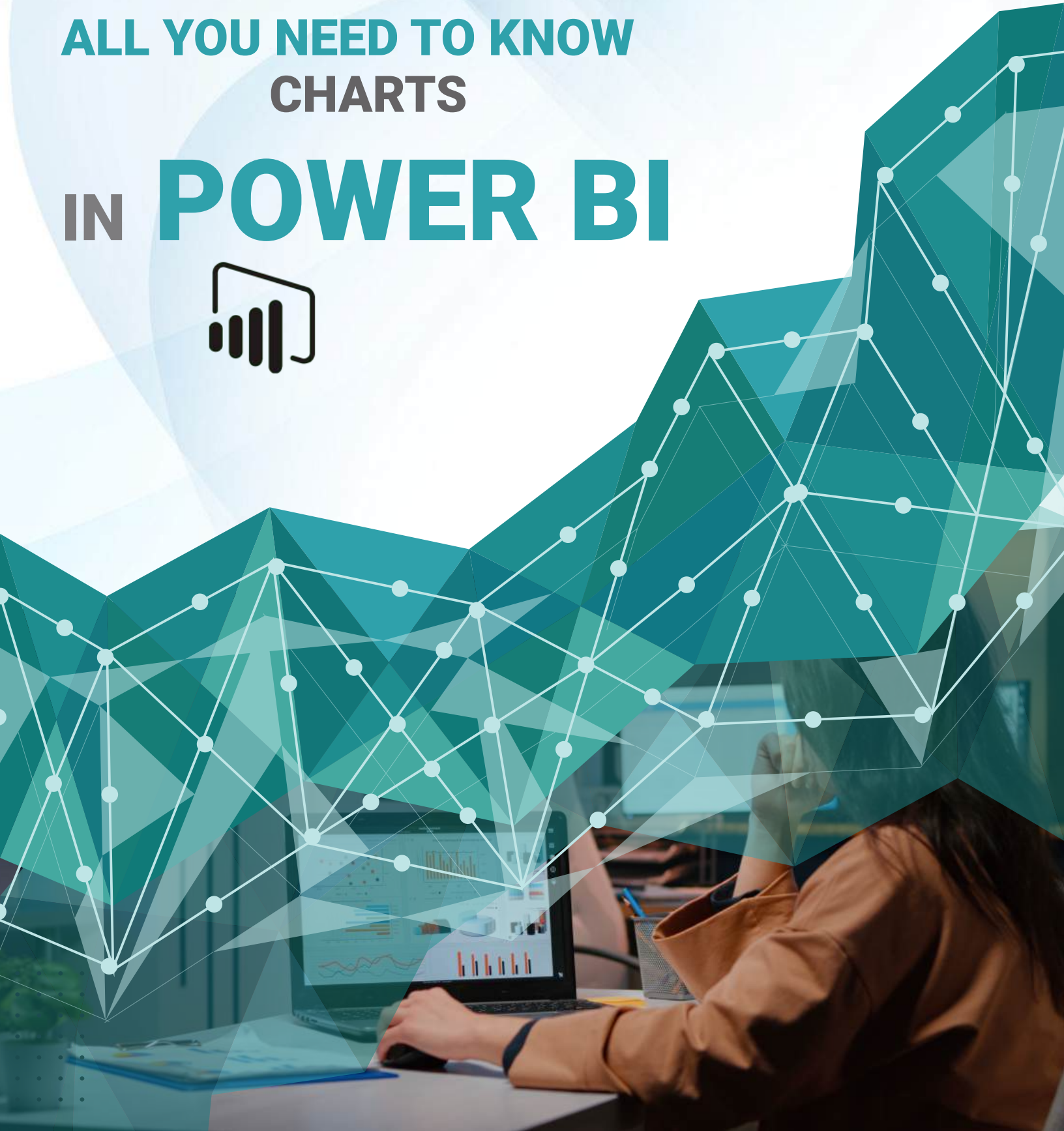
BRAINALYST
A Data Driven Company

BRAINALYST'S

ALL YOU NEED TO KNOW CHARTS IN POWER BI



ABOUT BRAINALYST

Brainalyst is a pioneering data-driven company dedicated to transforming data into actionable insights and innovative solutions. Founded on the principles of leveraging cutting-edge technology and advanced analytics, Brainalyst has become a beacon of excellence in the realms of data science, artificial intelligence, and machine learning.

OUR MISSION

At Brainalyst, our mission is to empower businesses and individuals by providing comprehensive data solutions that drive informed decision-making and foster innovation. We strive to bridge the gap between complex data and meaningful insights, enabling our clients to navigate the digital landscape with confidence and clarity.

WHAT WE OFFER

1. **Data Analytics and Consulting**

   Brainalyst offers a suite of data analytics services designed to help organizations harness the power of their data. Our consulting services include:

   - **Data Strategy Development:** Crafting customized data strategies aligned with your business objectives.
   - **Advanced Analytics Solutions:** Implementing predictive analytics, data mining, and statistical analysis to uncover valuable insights.
   - **Business Intelligence:** Developing intuitive dashboards and reports to visualize key metrics and performance indicators.

2. **Artificial Intelligence and Machine Learning**

   We specialize in deploying AI and ML solutions that enhance operational efficiency and drive innovation. Our offerings include:

   - **Machine Learning Models:** Building and deploying ML models for classification, regression, clustering, and more.
   - **Natural Language Processing:** Implementing NLP techniques for text analysis, sentiment analysis, and conversational AI.
   - **Computer Vision:** Developing computer vision applications for image recognition, object detection, and video analysis.

3. **Training and Development**

   Brainalyst is committed to fostering a culture of continuous learning and professional growth. We provide:

   - **Workshops and Seminars:** Hands-on training sessions on the latest trends and technologies in data science and AI.
   - **Online Courses:** Comprehensive courses covering fundamental to advanced topics in data analytics, machine learning, and AI.
   - **Customized Training Programs:** Tailored training solutions to meet the specific needs of organizations and individuals.

4. **Generative AI Solutions**

   As a leader in the field of Generative AI, Brainalyst offers innovative solutions that create new content and enhance creativity. Our services include:

   - **Content Generation:** Developing AI models for generating text, images, and audio.
   - **Creative AI Tools:** Building applications that support creative processes in writing, design, and media production.
   - **Generative Design:** Implementing AI-driven design tools for product development and optimization.

OUR JOURNEY

Brainalyst's journey began with a vision to revolutionize how data is utilized and understood. Founded by Nitin Sharma, a visionary in the field of data science, Brainalyst has grown from a small startup into a renowned company recognized for its expertise and innovation.

KEY MILESTONES:

- **Inception:** Brainalyst was founded with a mission to democratize access to advanced data analytics and AI technologies.
- **Expansion:** Our team expanded to include experts in various domains of data science, leading to the development of a diverse portfolio of services.
- **Innovation:** Brainalyst pioneered the integration of Generative AI into practical applications, setting new standards in the industry.
- **Recognition:** We have been acknowledged for our contributions to the field, earning accolades and partnerships with leading organizations.

  Throughout our journey, we have remained committed to excellence, integrity, and customer satisfaction. Our growth is a testament to the trust and support of our clients and the relentless dedication of our team.

WHY CHOOSE BRAINALYST?

Choosing Brainalyst means partnering with a company that is at the forefront of data-driven innovation. Our strengths lie in:

- **Expertise:** A team of seasoned professionals with deep knowledge and experience in data science and AI.
- **Innovation:** A commitment to exploring and implementing the latest advancements in technology.
- **Customer Focus:** A dedication to understanding and meeting the unique needs of each client.
- **Results:** Proven success in delivering impactful solutions that drive measurable outcomes.

JOIN US ON THIS JOURNEY TO HARNESS THE POWER OF DATA AND AI. WITH BRAINALYST, THE FUTURE IS DATA-DRIVEN AND LIMITLESS.

The Complete Guide to Power BI Visuals

Introduction

With out the right visuals, your Power BI report is redundant. To showcase powerful insights you need to understand when and how to use different visuals so that you can avoid wasting valuable time on building reports that don't make an impact.

What are Power BI Visuals

Visuals are simply a visual (picture) representation of your data, and are the most important part of any Power BI report as they are responsible for bringing your data to life. Visuals help you to tell a better data story, enabling your users to simply and easily identify and understand the patterns in your data.

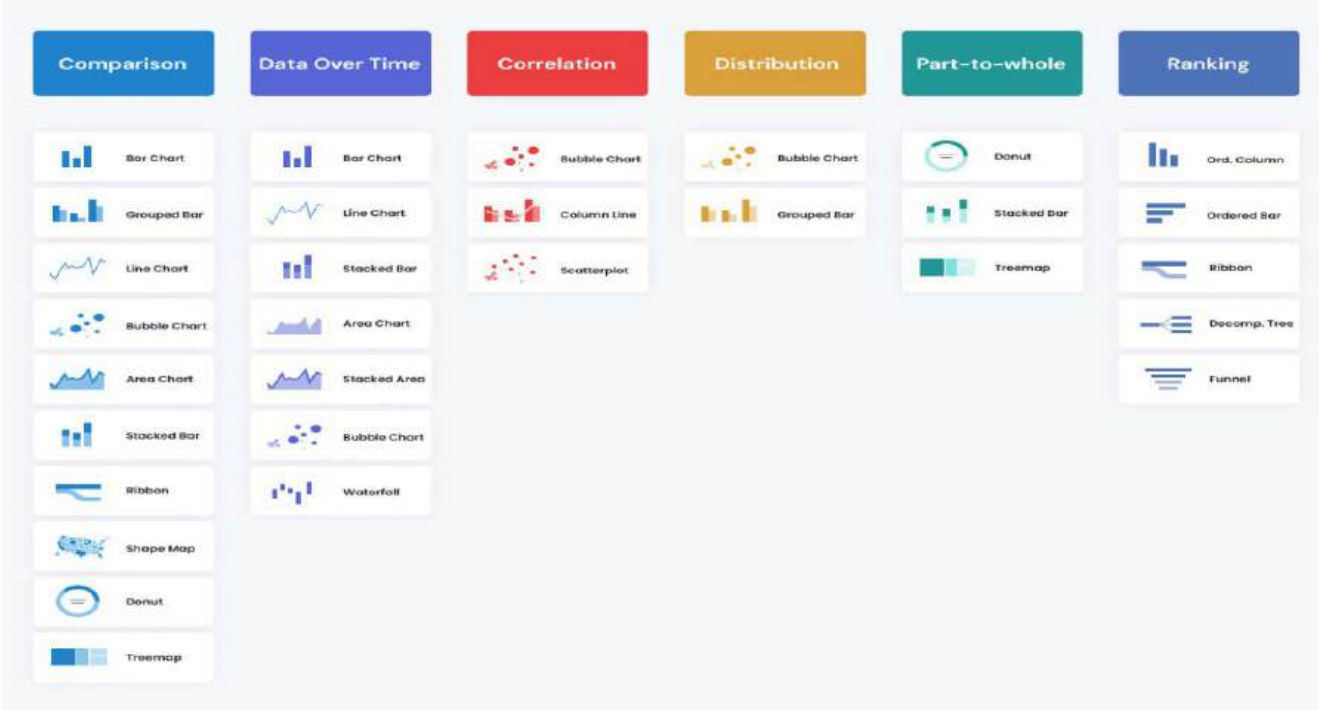
Types of Visuals

There are many ways to show your data through visualization.

When choosing your visual you need to consider what type of information your insight is looking to show.

The type of visual you chose to depict your data will dependon: the data you wish to communicate and what you want to say about that data.

Most visuals can be divided into the following 6 categori.



Brainalyst © 2021-2024

BAR CHART

When to Use a Bar Chart in Power BI

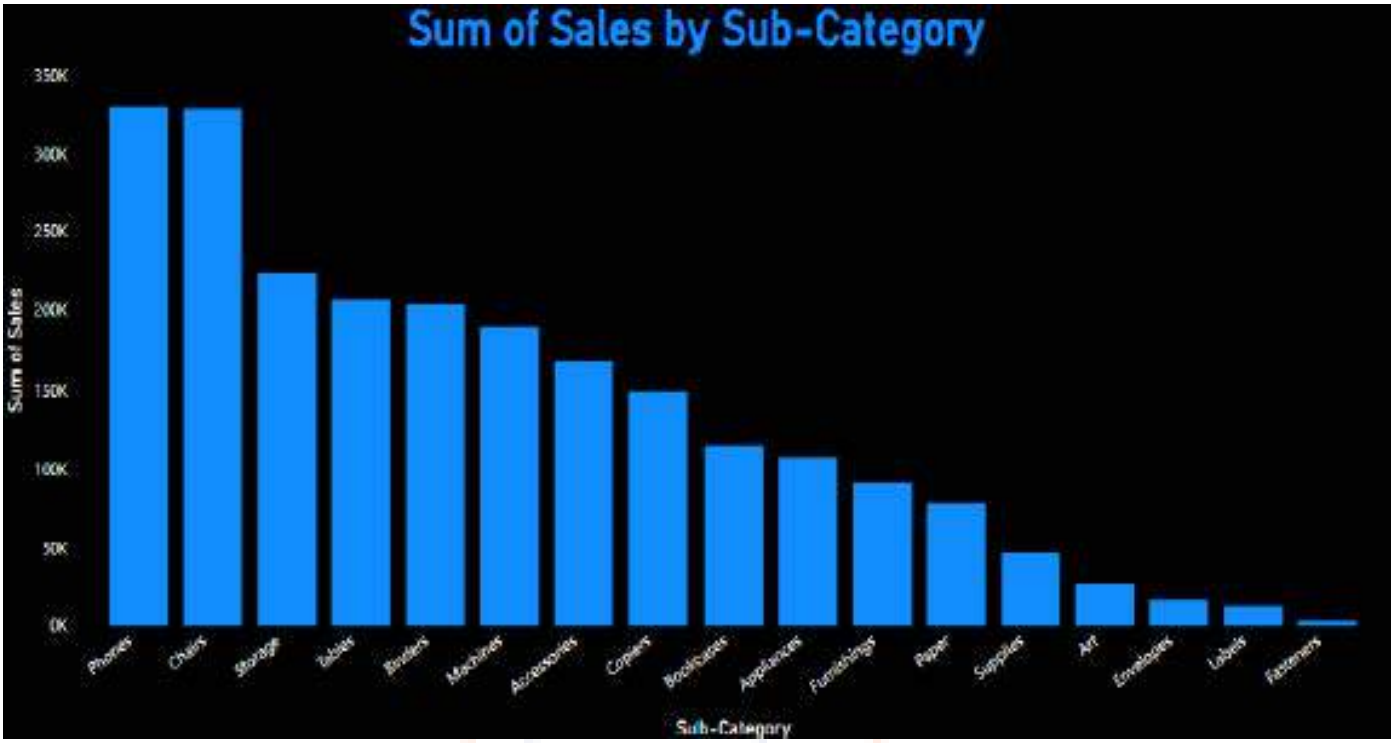
1. **Comparing Categorical Data**
    - Use a bar chart when you need to compare different categories or discrete groups of data.
    - It is ideal for showing data across categories where the order does not matter(e.g.,sales by region, product categories, etc.).
2. **Displaying Ranking and Order**
    - A bar chart is useful for displaying rankings or order of items (e.g., top-selling products, highest revenue-generating departments).
3. **Visualizing Changes Over Time**
    - When the changes over time involve discrete data points and are not continuous (e.g., annual sales by product).
4. **Highlighting Significant Data Points**
    - To emphasize significant data points or compare them against a benchmark or target.
5. **Showing Distribution**
    - Useful to show the distribution of data across different categories or groups.

Things to Keep in Mind

1. **Data Appropriateness**
    - Ensure that the data categories are discrete and non-continuous.
    - Use bar charts when the data categories are distinct and should not be mixed (e.g., do not mix product categories with time periods).
2. **Axis Configuration**
    - Use aclear and concise axislabeling.
    - The x-axis typically represents the categories, and the y-axis represents the values.
    - Ensure the scales are appropriate and not misleading (e.g., avoid starting the y-axis at a value other than zero unless justified).
3. **Bar Spacing**
    - Maintain consistent spacing between bars to avoid confusion.
    - Ensure the bars are evenly spaced and not too close or too far apart.

4. **Bar Orientation**
    - Decide whether to use horizontal or vertical bars:
        - Vertical bars are usually better for time-series data.
        - Horizontal bars are better for long category names or when comparing many categories.
5. **Color Usage**
    - Use distinct and consistent colors for different categories.
    - Avoid using too many colors that can cause confusion. Utilize a color legend if necessary.
6. **Data Labels**
    - Include data labels to provide precise values for each bar.
    - Ensure labels are readable and do not overlap.
7. **Title and Description**
    - Provide a clear and descriptive title.
    - Include additional descriptions if needed to clarify what the chart represents.
8. **Sorting**
    - Sort the bars in a meaningful order (e.g., descending order for rankings).
    - Alphabetical sorting can be used if the data categories do not have a natural order.
9. **Comparison with Other Charts**
    - Compare and decide if a bar chart is the best option or if another type (e.g., line chart, pie chart) would be more effective for the data representation.
10. **Interaction and Drill-Down**
    - Utilize Power BI's interactive features to allow users to drill down into more detailed data if needed.
    - Enable tooltips to provide additional context or information when users hover over the bars.

Brainalyst © 2021-2024



BRAINALYST
A Data Driven Company

STACKED BAR CHART

When to Use a Stacked Bar Chart in Power BI

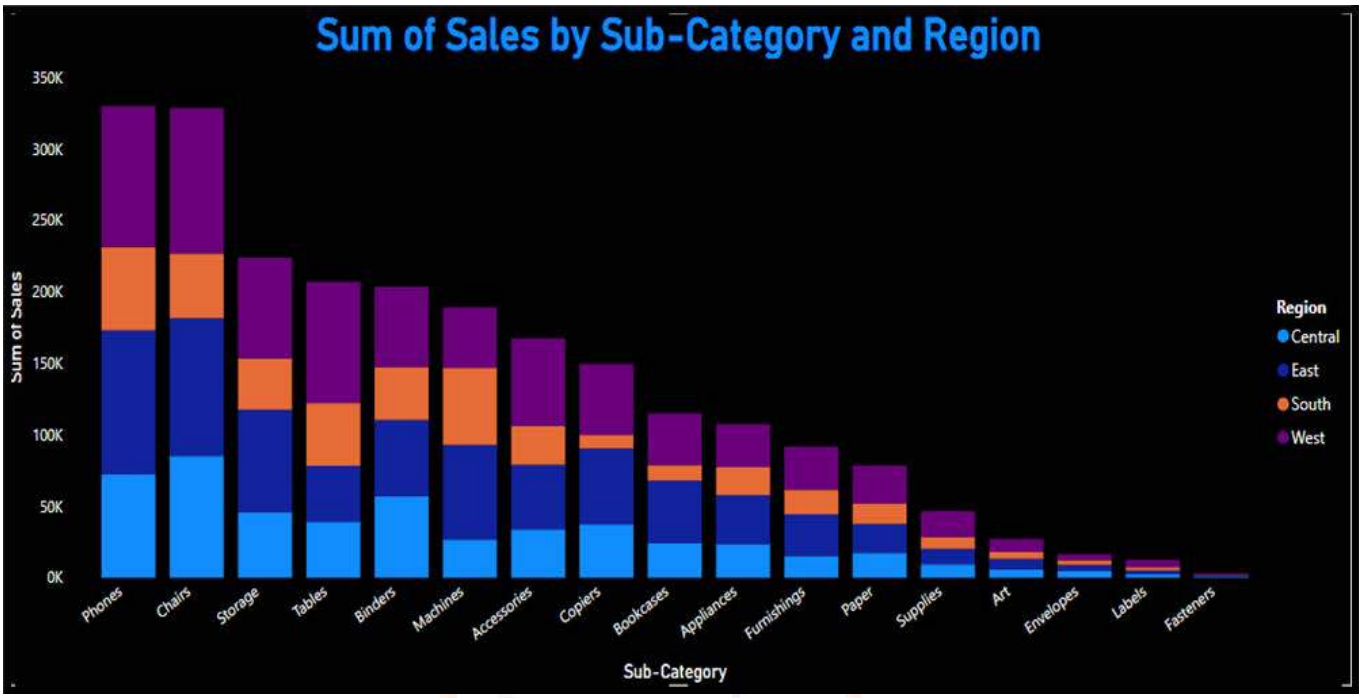
1. **Comparing Multiple Categories within a Group**
   - Use a stacked bar chart to compare different categories within a single group (e.g., sales by product within each region).
   - It is ideal for showing the composition of data across different segments of a category.
2. **Showing Part-to-Whole Relationships**
   - Useful for displaying how individual parts contribute to a whole (e.g., market share of different products within total sales).
3. **Highlighting Cumulative Data**
   - To highlight cumulative data and understand the total value along with the breakdown of individual components.
4. **Visualizing Changes Over Time with Categories**
   - When you need to show changes over time and compare multiple categories within each time period (e.g., quarterly revenue by product line).

Things to Keep in Mind

1. **Data Structure**
   - Ensure that the data is structured properly with distinct categories that sum up to a meaningful total.
   - Verify that each bar segment represents a part of the whole and adds up correctly.
2. **Color Coding**
   - Use consistent and distinguishable colors for different segments.
   - Make sure the color legend is clear and easy to understand.
3. **Readability of Segments**
   - Ensure that smaller segments are readable and not too thin.
   - Consider using data labels or tooltips to provide precise values for each segment.
4. **Axis Configuration**
   - Label the axes clearly, with the x-axis representing the categories and the y-axis representing the values.
   - Use an appropriate scale to ensure that the relative sizes of the segments are easily comparable.

Brainalyst © 2021-2024

5. **Bar Orientation**
    - Decide whether to use horizontal or vertical stacked bars:
        - Horizontal stacked bars are often better for categories with longer names.
        - Vertical stacked bars work well for time-series data or where the y-axis represents a numerical value.
6. **Sorting and Ordering**
    - Sort the categories in a meaningful order, such as descending order of total value.
    - Consider the order of the segments within each bar for better visual clarity.
7. **Title and Description**
    - Provide a clear and descriptive title.
    - Include additional explanations if necessary to clarify what each segment represents.
8. **Data Labels**
    - Include data labels for both the total value and individual segments if possible.
    - Ensure that labels are positioned clearly and do not overlap.
9. **Comparison with Other Charts**
    - Evaluate if a stacked bar chart is the best option or if another type (e.g., clustered bar chart, stacked column chart) would be more suitable.
    - Consider the effectiveness of showing both individual segment values and the total value.
10. **Interactive Features**
    - Utilize Power BI's interactive features to allow users to filter and drill down into more detailed data.
    - Enable tooltips to provide additional context or information when users hover over segments.
11. **Visibility of Small Segments**
    - Pay attention to small segments that might be difficult to see.
    - Consider alternative ways to highlight these segments if they are important (e.g., using annotations or callouts).



BRAINALYST
A Data Driven Company

Brainalyst © 2021-2024

BUBBLE CHART

When to Use a Bubble Chart in Power BI

1. **Visualizing Relationships Between Three Variables**

   Use a bubble chart when you need to visualize the relationship between three different variables, where the x-axis, y-axis, and bubble size each represent a different variable.

2. **Highlighting Data Density**

   - Useful for highlighting the density of data points and identifying clusters or patterns within the data.

3. **Comparing Data Points with a Third Dimension**

   - Ideal for comparing data points where an additional variable is represented by the size of the bubbles (e.g., sales revenue by region and product with bubble size representing market share).

4. **Showing Distribution and Outliers**

   - Helps to show the distribution of data points and identify any outliers or anomalies.

Things to Keep in Mind

1. **Choosing Variables Appropriately**

   - Ensure the variables chosen for the x-axis, y-axis, and bubble size are appropriate and relevant to the analysis.
   - The bubble size should represent a meaningful quantitative variable.

2. **Scale and Axis Configuration**

   - Use a clear and consistent scale for both the x-axis and y-axis.
   - Label the axes clearly and ensure the units are easily understandable.

3. **Bubble Size Clarity**

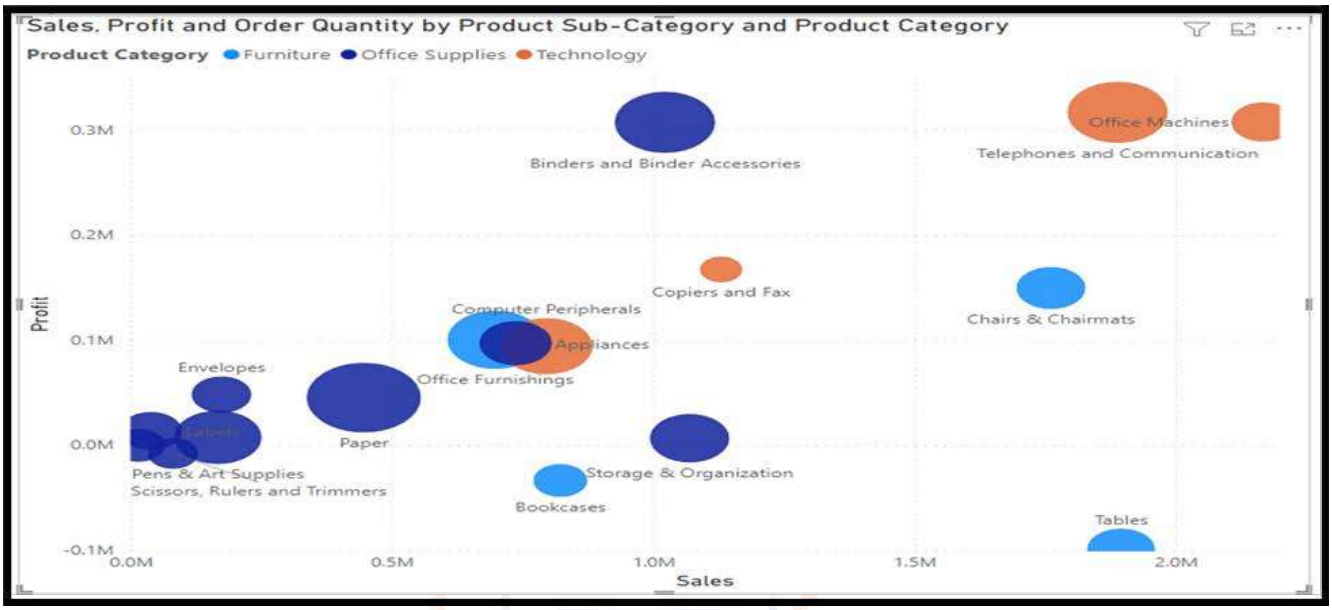
   - Ensure that the size of the bubbles is proportional to the values they represent.
   - Avoid using too many small bubbles that can be difficult to distinguish.

4. **Color Coding**

   - Use different colors to represent different categories or groups if necessary.
   - Ensure that the color legend is clear and easy to understand.

5. **Avoiding Overcrowding**
    - Be mindful of overcrowding when there are too many data points.
    - Consider using filters or zoom features to focus on specific areas of interest.
6. **Interactivity and Tooltips**
    - Utilize Power BI's interactive features to allow users to filter data and drill down into more detailed information.
    - Enable tooltips to provide additional context or information when users hover over bubbles.
7. **Bubble Overlap**
    - Be aware of overlapping bubbles which can obscure data points.
    - Use transparency or outline features to make overlapping bubbles distinguishable.
8. **Data Labels**
    - Include data labels or tooltips to provide precise values for each bubble.
    - Ensure labels are readable and do not clutter the chart.
9. **Title and Description**
    - Provide a clear and descriptive title.
    - Include additional descriptions if needed to clarify what each axis and bubble size represent.
10. **Comparison with Other Charts**
    - Consider if a bubble chart is the best option or if another type (e.g., scatter plot with varying point sizes, heat map) would be more effective.
    - Evaluate the effectiveness of showing the relationship and distribution with three dimensions.
11. **Handling Outliers**
    - Identify and handle outliers appropriately, as they can disproportionately affect the bubble sizes and readability.
    - Consider using a logarithmic scale if there is a large range of values.

Brainalyst © 2021-2024


Sales, Profit and Order Quantity by Product Sub-Category and Product Category
Product Category
Furniture
Office Supplies
Technology
Profit
0.3M
0.2M
0.1M
0.0M
-0.1M
0.0M
0.5M
1.0M
1.5M
2.0M
Sales
Binders and Binder Accessories
Office Machines
Telephones and Communication
Copiers and Fax
Chairs & Chairmats
Computer Peripherals
Appliances
Office Furnishings
Envelopes
Paper
Labels
Pens & Art Supplies
Scissors, Rulers and Trimmers
Storage & Organization
Bookcases
Tables

TREE MAP

When to Use a Tree Map in Power BI

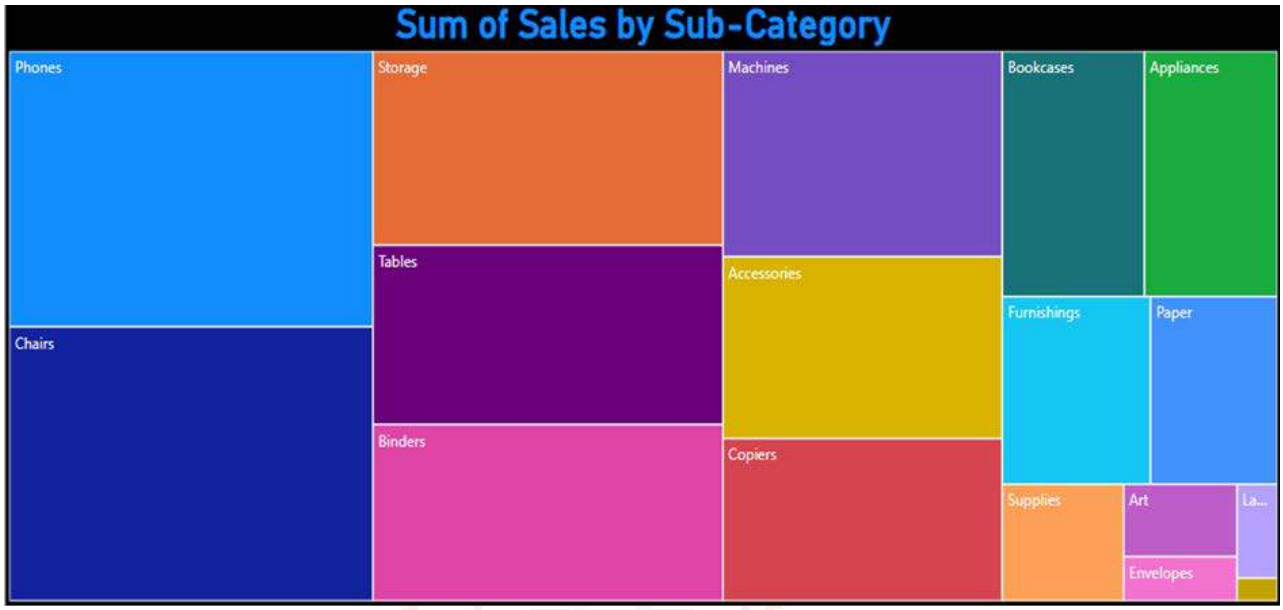
1. **Displaying Hierarchical Data**
   - Use a tree map to display hierarchical data and show the relationship between parts and the whole.
2. **Visualizing Proportions**
   - Ideal for visualizing proportions and how individual segments contribute to the total.
3. **Highlighting Large Data Sets**
   - Useful for summarizing large data sets and making them more understandable by breaking them down into smaller segments.
4. **Comparing Categories**
   - Effective for comparing the size of different categories within a hierarchy.
5. **Analyzing Parts of a Whole**
   - Perfect for analyzing how different parts make up a whole, such as sales by region and product.

Things to Keep in Mind

1. **Data Hierarchy Structure**
   - Ensure that your data is structured in a hierarchical manner.
   - The tree map should represent nested categories within broader categories.
2. **Color Coding**
   - Use distinct colors to represent different categories or levels in the hierarchy.
   - Ensure that the color legend is clear and easy to understand.
3. **Label Clarity**
   - Include labels for each segment to provide context and clarity.
   - Ensure that labels are readable and do not overlap or clutter the map.
4. **Size Representation**
   - Make sure the size of each rectangle accurately represents the quantitative value it corresponds to.
   - Larger rectangles should represent larger values.

Brainalyst © 2021-2024

5. **Depth of Hierarchy**
    - Be mindful of the depth of hierarchy displayed. Too many levels can make the tree map cluttered and hard to interpret.
    - Use filters or drill-down features to manage deeper levels of hierarchy effectively.
6. **Interactivity and Drill-Down**
    - Utilize Power BI's interactive features to allow users to drill down into more detailed data within the hierarchy.
    - Enable tooltips to provide additional information when users hover over different segments.
7. **Color Consistency**
    - Maintain consistent color usage for similar categories across different tree maps to help users recognize patterns and categories quickly.
8. **Comparison with Other Charts**
    - Evaluate if a tree map is the best option or if another type (e.g., pie chart, bar chart) would be more suitable for showing part-to-whole relationships.
    - Consider the effectiveness of displaying hierarchical data with other visualization types.
9. **Title and Description**
    - Provide a clear and descriptive title.
    - Include additional descriptions if necessary to clarify what each level of the hierarchy represents.
10. **Handling Small Segments**
    - Be aware of very small segments that might be hard to see and interpret.
    - Consider grouping smaller segments together or using a different visualization if they are numerous.
11. **Data Labels and Tooltips**
    - Include data labels or tooltips to provide precise values and additional context for each segment.
    - Ensure that the labels and tooltips are helpful and not overly complex.



BRAINALYST
A Data Driven Company

Brainalyst © 2021-2024

GROUPED BAR

When to Use a Grouped Bar Chart in Power BI

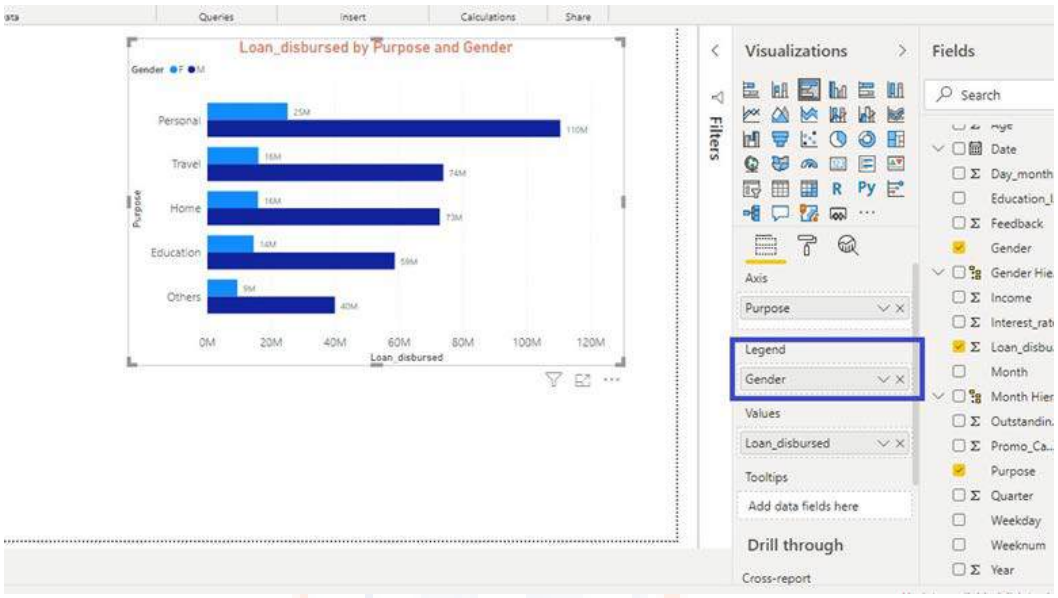
1. **Comparing Multiple Categories Across Different Groups**
   - Use a grouped bar chart when you need to compare multiple categories within different groups or sub-categories.
   - Ideal for showing side-by-side comparisons of data for different groups (e.g., sales by product category across different regions).
2. **Showing Detailed Comparisons**
   - Effective for detailed comparisons between categories within the same group, allowing for an easy visual comparison of each category.
3. **Analyzing Trends Over Time**
   - Useful for analyzing trends over time when you have multiple categories to compare within each time period.
4. **Highlighting Variability**
   - Great for highlighting variability between different categories within the same group.

Things to Keep in Mind

1. **Category and Group Selection**
   - Ensure the categories and groups chosen are relevant and meaningful.
   - Each group should have the same set of categories to allow for effective comparisons.
2. **Color Coding**
   - Use distinct colors for different categories within each group.
   - Maintain a consistent color scheme to avoid confusion.
3. **Axis Configuration**
   - Label the axes clearly, with the x-axis typically representing the groups and the y-axis representing the values.
   - Use an appropriate scale to ensure accurate representation of the data.
4. **Bar Spacing and Width**
   - Maintain consistent spacing between bars within a group and between different groups.
   - Ensure that bars are not too narrow, making them difficult to interpret.

5. **Legend and Labels**
    - Include a legend to explain the colors representing different categories.
    - Use clear and concise labels for each bar to provide context.
6. **Title and Description**
    - Provide a clear and descriptive title.
    - Include additional descriptions if needed to clarify what each group and category represent.
7. **Data Labels**
    - Include data labels to provide precise values for each bar.
    - Ensure labels are readable and do not overlap.
8. **Sorting and Ordering**
    - Sort the groups in a meaningful order (e.g., chronological order for time-series data).
    - Consider the order of categories within each group to enhance readability.
9. **Comparison with Other Charts**
    - Evaluate if a grouped bar chart is the best option or if another type (e.g., stacked bar chart, line chart) would be more effective for the data comparison.
    - Consider the clarity and simplicity of the visualization.
10. **Interactivity and Drill-Down**
    - Utilize Power BI's interactive features to allow users to filter and drill down into more detailed data.
    - Enable tooltips to provide additional context or information when users hover over bars.
11. **Handling Large Data Sets**
    - Be cautious when dealing with large data sets, as too many bars can make the chart cluttered and hard to read.
    - Consider using filters or aggregating data to manage the complexity.
12. **Comparison Clarity**
    - Ensure that the chart effectively highlights the differences and similarities between categories within each group.
    - Use visual aids like gridlines or reference lines to enhance comparison.

Brainalyst © 2021-2024



BRAINALYST
A Data Driven Company

RIBBON CHART

When to Use a Ribbon Chart in Power BI

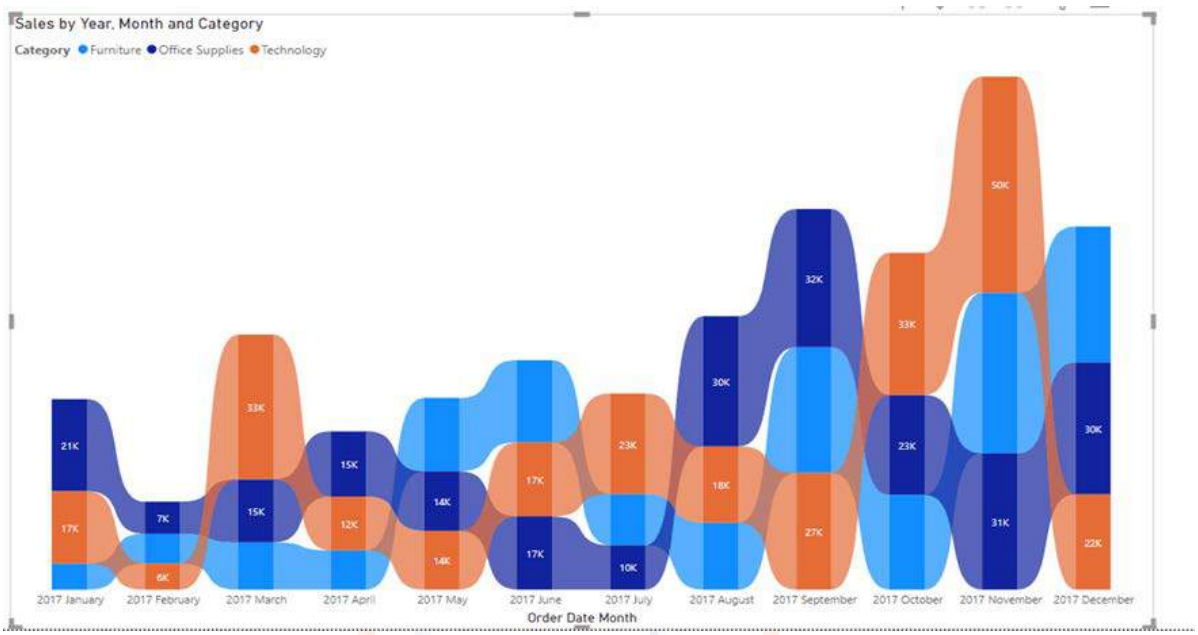
1. **Showing Ranking Over Time**
   - Use a ribbon chart to display changes in rankings over time or across different categories. It is particularly effective for visualizing how the ranking of items changes across different periods or segments.
2. **Visualizing Multiple Categories**
   - Ideal for comparing multiple categories and understanding how their values and rankings evolve over time.
3. **Highlighting Trends and Patterns**
   - Useful for highlighting trends, patterns, and shifts in data across categories and time.
4. **Analyzing Part-to-Whole Relationships**
   - Great for analyzing how individual categories contribute to the total and how these contributions change over time.

Things to Keep in Mind

1. **Data Appropriateness**
   - Ensure that the data is suitable for showing rankings over time or across categories.
   - The data should have multiple categories with values that change over the dimension you are analyzing (e.g., time, regions).
2. **Color Coding**
   - Use distinct colors for different categories to differentiate between them easily.
   - Maintain consistent color usage throughout the chart to help users track categories across the chart.
3. **Labeling**
   - Include clear and concise labels for each ribbon and axis.
   - Ensure that labels for each category are easily readable and do not overlap.
4. **Axis Configuration**
   - Label the axes clearly, with the x-axis typically representing the time periods or categories and the y-axis representing the values or rankings.
   - Use an appropriate scale to ensure the data is represented accurately and clearly.

Brainalyst © 2021-2024

5. **Ribbon Width**
    - Ensure that the width of the ribbons accurately represents the values they correspond to.
    - Avoid overly narrow ribbons that can make the chart difficult to interpret.
6. **Sorting and Ordering**
    - Sort the categories in a meaningful order to enhance the readability of the chart.
    - Consider the most logical order for your data, such as chronological order for time-series data.
7. **Title and Description**
    - Provide a clear and descriptive title.
    - Include additional descriptions if needed to clarify what each axis and ribbon represent.
8. **Interactivity and Tooltips**
    - Utilize Power BI's interactive features to allow users to filter and drill down into more detailed data.
    - Enable tooltips to provide additional context or information when users hover over ribbons.
9. **Legend and Guide**
    - Include a legend to explain the colors and what each ribbon represents.
    - Ensure the legend is positioned in a way that does not clutter the chart.
10. **Comparison with Other Charts**
    - Evaluate if a ribbon chart is the best option or if another type (e.g., line chart, stacked area chart) would be more effective for the data analysis.
    - Consider how well the ribbon chart communicates the changes in rankings and contributions.
11. **Highlighting Key Changes**
    - Use visual aids like annotations or callouts to highlight significant changes or trends in the data.
    - Make sure important shifts and trends are easily noticeable.
12. **Handling Complexity**
    - Be cautious when dealing with a large number of categories, as the chart can become cluttered.
    - Consider aggregating data or using filters to manage the complexity and maintain readability.



BRAINALYST
A Data Driven Company

AREA CHART

When to Use an Area Chart in Power BI

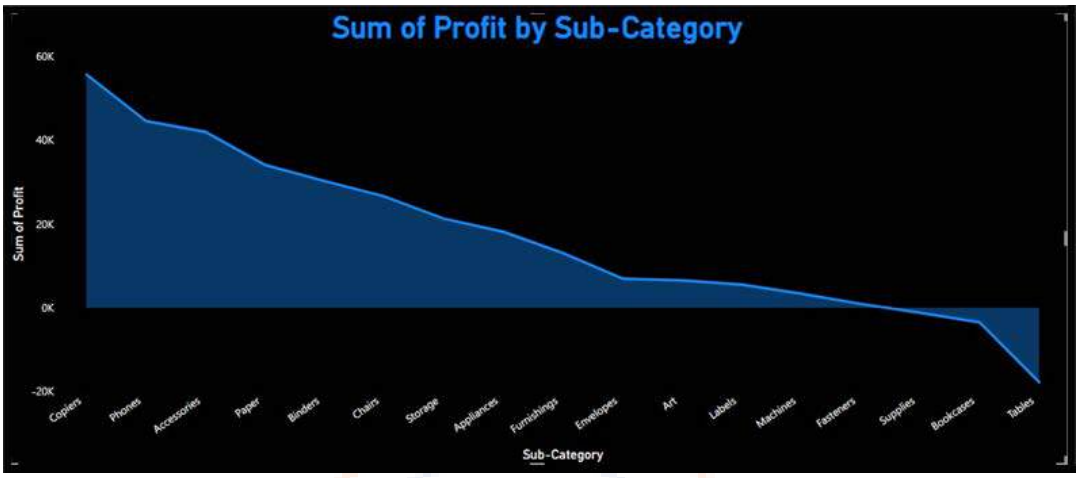
1. **Displaying Trends Over Time**
   - Use an area chart to show how values change over time. It is particularly effective for visualizing trends and patterns over a continuous time period.
2. **Comparing Multiple Categories Over Time**
   - Ideal for comparing multiple categories and understanding how their contributions evolve over time.
3. **Visualizing Part-to-Whole Relationships**
   - Useful for showing how individual segments contribute to the total and how these contributions change over time.
4. **Highlighting Cumulative Data**
   - Effective for displaying cumulative data and understanding the overall trend along with individual segment contributions.

Things to Keep in Mind

1. **Data Appropriateness**
   - Ensure that the data is suitable for continuous time series analysis.
   - The data should have a clear time or sequence dimension to represent on the x-axis.
2. **Stacked vs. Non-Stacked Area Charts**
   - Decide between a stacked area chart and a non-stacked area chart:
     - Stacked Area Chart: Use when you want to show the contribution of each category to the total over time.
     - Non-Stacked Area Chart: Use when comparing trends of different categories without focusing on the total.
3. **Color Coding**
   - Use distinct and consistent colors for different categories to differentiate them easily.
   - Ensure the color legend is clear and easy to understand.
4. **Labeling and Axes Configuration**
   - Include clear and concise labels for the x-axis (typically time) and y-axis (values).
   - Make sure the axes are appropriately scaled to represent the data accurately.

5. **Transparency and Overlap**
    - Use transparency to handle overlapping areas, especially in non-stacked area charts, so that all categories remain visible.
    - Ensure that overlapping areas do not obscure important data points.
6. **Title and Description**
    - Provide a clear and descriptive title.
    - Include additional descriptions if necessary to clarify what each axis and area represent.
7. **Legend and Guide**
    - Include a legend to explain the colors and what each area represents.
    - Position the legend in a way that does not clutter the chart.
8. **Interactivity and Tooltips**
    - Utilize Power BI's interactive features to allow users to filter and drill down into more detailed data.
    - Enable tooltips to provide additional context or information when users hover over areas.
9. **Highlighting Key Trends**
    - Use visual aids like annotations or callouts to highlight significant trends or changes in the data.
    - Ensure important trends and changes are easily noticeable.
10. **Data Labels**
    - Include data labels to provide precise values for important points on the chart.
    - Ensure labels are readable and do not overlap.
11. **Handling Complexity**
    - Be cautious when dealing with a large number of categories, as the chart can become cluttered.
    - Consider using filters or aggregating data to manage complexity and maintain readability.
12. **Comparison with Other Charts**
    - Evaluate if an area chart is the best option or if another type (e.g., line chart, bar chart) would be more effective for the data analysis.
    - Consider how well the area chart communicates trends and part-to-whole relationships.

Brainalyst © 2021-2024


Sum of Profit by Sub-Category
60K
40K
20K
0K
-20K
Sum of Profit
Copiers
Phones
Accessories
Paper
Binders
Chairs
Storage
Appliances
Furnishings
Envelopes
Art
Labels
Machines
Fasteners
Supplies
Bookcases
Tables
Sub-Category

LINE CHART

When to Use a Line Chart in Power BI

1. **Displaying Trends Over Time**
   - Use a line chart to show how values change over continuous time periods. It is particularly effective for visualizing trends and patterns over time.
2. **Comparing Multiple Data Series**
   - Ideal for comparing multiple data series to understand how different variables interact over time.
3. **Highlighting Data Continuity**
   - Useful for emphasizing the continuity of data points and showing how data changes in a connected manner.
4. **Identifying Patterns and Outliers**
   - Effective for identifying patterns, trends, and outliers within your data.

Things to Keep in Mind

1. **Data Appropriateness**
   - Ensure that your data is continuous and suitable for time series analysis.
   - Line charts are best for data with a natural order or progression, typically over time.
2. **Axes Configuration**
   - Label the x-axis clearly, typically representing time or sequential data.
   - Label the y-axis with the values being measured.
   - Ensure the axes are appropriately scaled to accurately represent the data.
3. **Multiple Series**
   - Use different lines to represent multiple data series.
   - Ensure each line is clearly distinguishable, typically through the use of different colors or line styles.
4. **Color Coding**
   - Use distinct colors for different lines to differentiate them easily.
   - Maintain a consistent color scheme to help users track the data series across the chart.

Brainalyst © 2021-2024

5. **Data Points and Markers**
    - Include markers for individual data points if necessary to highlight specific values.
    - Ensure that markers do not clutter the chart, especially with dense data.
6. **Line Style and Thickness**
    - Choose appropriate line styles and thicknesses to ensure clarity and readability.
    - Use dashed or dotted lines for emphasis or differentiation if needed.
7. **Title and Description**
    - Provide a clear and descriptive title.
    - Include additional descriptions if necessary to clarify what each axis and line represent.
8. **Legend and Guide**
    - Include a legend to explain the colors and styles of the lines.
    - Position the legend in a way that does not clutter the chart.
9. **Interactivity and Tooltips**
    - Utilize Power BI's interactive features to allow users to filter and drill down into more detailed data.
    - Enable tooltips to provide additional context or information when users hover over lines or data points.
10. **Highlighting Key Trends**
    - Use visual aids like annotations or callouts to highlight significant trends or changes in the data.
    - Ensure important trends and changes are easily noticeable.
11. **Data Labels**
    - Include data labels to provide precise values for important points on the chart.
    - Ensure labels are readable and do not overlap.
12. **Handling Complexity**
    - Be cautious when dealing with a large number of data series, as the chart can become cluttered.
    - Consider using filters or aggregating data to manage complexity and maintain readability.

13. Comparison with Other Charts

- Evaluate if a line chart is the best option or if another type (e.g., area chart, bar chart) would be more effective for the data analysis.
- Consider how well the line chart communicates trends, patterns, and relationships between data series.

BRAINALYST
A Data Driven Company

SHAPE MAP CHART

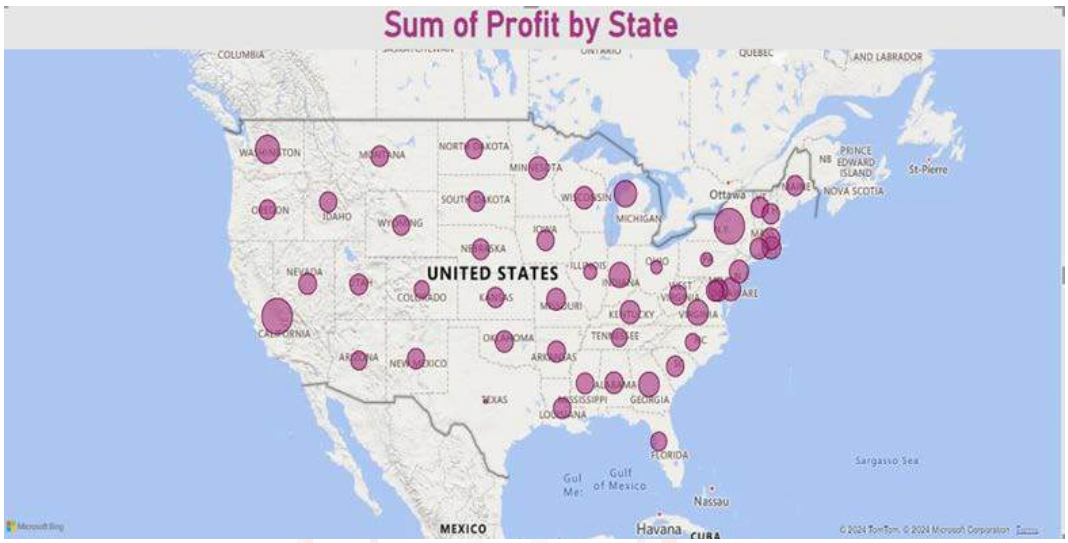
When to Use a Shape Map Chart in Power BI

1. **Geographical Data Visualization**
   - Use a shape map chart to display data related to geographical regions. It is particularly effective for visualizing metrics across different geographic areas.
2. **Comparing Regional Data**
   - Ideal for comparing data across various regions, such as states, countries, or any custom-defined geographic boundaries.
3. **Highlighting Spatial Patterns**
   - Useful for highlighting patterns and trends that have a spatial component, helping to identify geographic clusters or outliers.
4. **Visualizing Distribution**
   - Effective for showing the distribution of values across a map, providing a clear view of how data varies by location.

Things to Keep in Mind

1. **Geographical Boundaries and Data**
   - Ensure your data includes geographic identifiers (e.g., country names, state codes) that match the map's boundaries.
   - Verify that the shape map's geographic regions accurately represent your data.
2. **Color Coding and Gradients**
   - Use a color gradient to represent different values, with a clear legend explaining what each color or shade represents.
   - Choose colors that are easily distinguishable and consider colorblind-friendly palettes.
3. **Scale and Legend Configuration**
   - Include a legend that explains the color scale and what each color represents.
   - Ensure the legend is positioned in a way that does not obstruct the map.
4. **Data Normalization**
   - Normalize your data if necessary to account for differences in region sizes or populations.
   - Consider using per capita values or percentages to provide a more accurate comparison.

5. **Interactivity and Tooltips**
   - Utilize Power BI's interactive features to allow users to zoom in and out, filter data, and explore specific regions in detail.
   - Enable tooltips to provide additional context or information when users hover over different regions.
6. **Title and Description**
   - Provide a clear and descriptive title.
   - Include additional descriptions if necessary to clarify what the map represents and what the color gradient indicates.
7. **Boundary Clarity**
   - Ensure that the boundaries of regions are clearly visible and distinct.
   - Consider using boundary lines or different shades to differentiate adjacent regions.
8. **Data Labels**
   - Include data labels or markers for important regions to provide precise values.
   - Ensure labels are readable and do not overlap or clutter the map.
9. **Handling Missing Data**
   - Address missing data appropriately, either by using a specific color to indicate missing values or by providing an explanatory note.
   - Ensure that regions with missing data are clearly identifiable.
10. **Comparison with Other Charts**
    - Evaluate if a shape map chart is the best option or if another type (e.g., choropleth map, bubble map) would be more effective for your data visualization.
    - Consider how well the shape map chart communicates spatial patterns and regional comparisons.
11. **Data Accuracy**
    - Ensure that your geographic data is accurate and up to date.
    - Double-check geographic boundaries and region names to avoid misrepresentation.
12. **Map Customization**
    - Customize the map to highlight specific regions or data points if needed.
    - Consider using different map projections if the default one does not suit your data well.



BRAINALYST
A Data Driven Company

DONUT CHART

When to Use a Donut Chart in Power BI

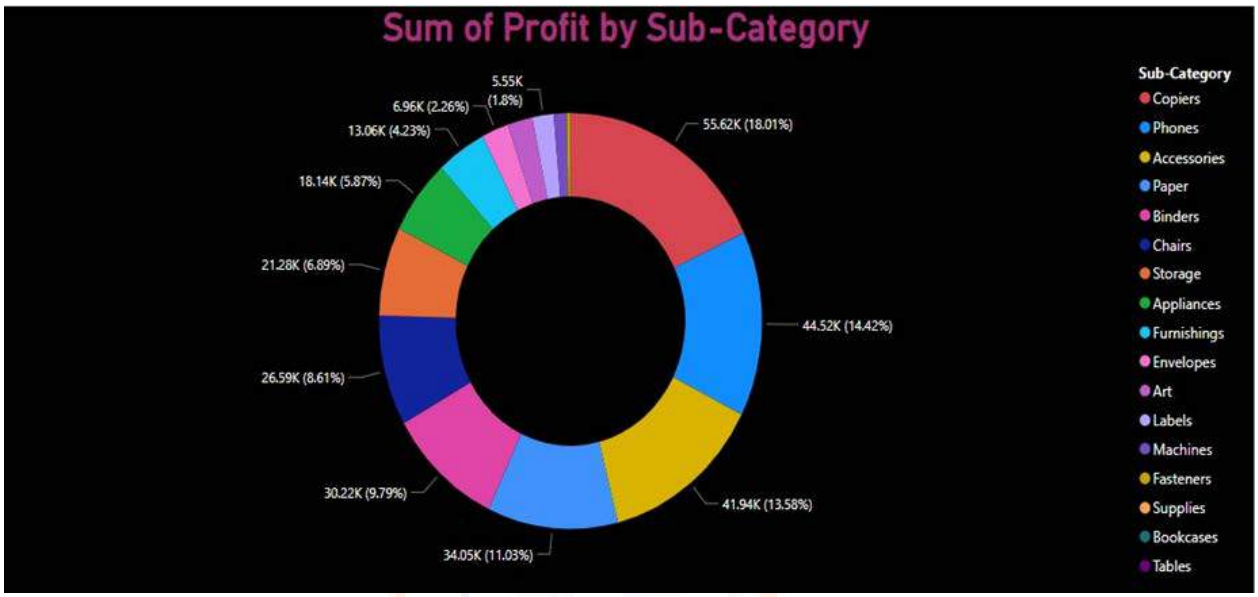
1. **Showing Parts of a Whole**
   - Use a donut chart to visualize how different segments contribute to the total. It is effective for displaying proportions and percentages.
2. **Comparing Categories**
   - Ideal for comparing different categories within a dataset and understanding their relative sizes.
3. **Highlighting Key Segments**
   - Useful for emphasizing one or more key segments within the whole, especially with the central empty space to add additional information.
4. **Visual Appeal**
   - Effective for creating visually appealing reports where the circular design can enhance the aesthetic quality.

Things to Keep in Mind

1. **Segment Count**
   - Limit the number of segments to avoid clutter and ensure readability. Too many segments can make the chart difficult to interpret.
   - Aim for no more than 5-7 segments for optimal clarity.
2. **Color Coding**
   - Use distinct and contrasting colors for different segments to differentiate them easily.
   - Ensure that the color legend is clear and matches the segments accurately.
3. **Labeling and Percentages**
   - Include labels or data labels for each segment to show the actual values or percentages.
   - Ensure labels are readable and do not overlap. Consider using callouts for better clarity.
4. **Central Space Utilization**
   - Utilize the central space effectively, often for placing the total value, a key metric, or an explanatory label.
   - Ensure the central information is concise and relevant.

Brainalyst © 2021-2024

5. **Comparison with Pie Charts**
    - Evaluate if a donut chart is more appropriate than a pie chart. Donut charts are generally preferred when you want to reduce the visual dominance of the whole or when you have a meaningful way to use the central space.
6. **Visual Hierarchy**
    - Emphasize key segments by using brighter colors or larger segments to draw attention.
    - Use consistent color schemes across similar charts for better comparison.
7. **Title and Description**
    - Provide a clear and descriptive title.
    - Include additional descriptions if necessary to clarify what each segment represents.
8. **Interactivity and Tooltips**
    - Utilize Power BI's interactive features to allow users to drill down into more detailed data.
    - Enable tooltips to provide additional context or information when users hover over segments.
9. **Data Accuracy**
    - Ensure that the segment sizes accurately reflect the data proportions.
    - Double-check calculations for percentages and ensure the total adds up to 100%.
10. **Handling Small Segments**
    - Be mindful of very small segments that may be hard to see or click on.
    - Consider grouping small segments into an "Other" category for better readability.
11. **Comparison with Other Charts**
    - Evaluate if a donut chart is the best option or if another type (e.g., bar chart, stacked bar chart) would be more effective for your data analysis.
    - Consider how well the donut chart communicates the part-to-whole relationship and segment comparisons.
12. **Avoiding Redundancy**
    - Avoid using a donut chart if the data can be better represented with a simpler chart type.
    - Ensure that the donut chart adds value and insight to the data presentation.



BRAINALYST
A Data Driven Company

FUNNEL CHART

When to Use a Funnel Chart in Power BI

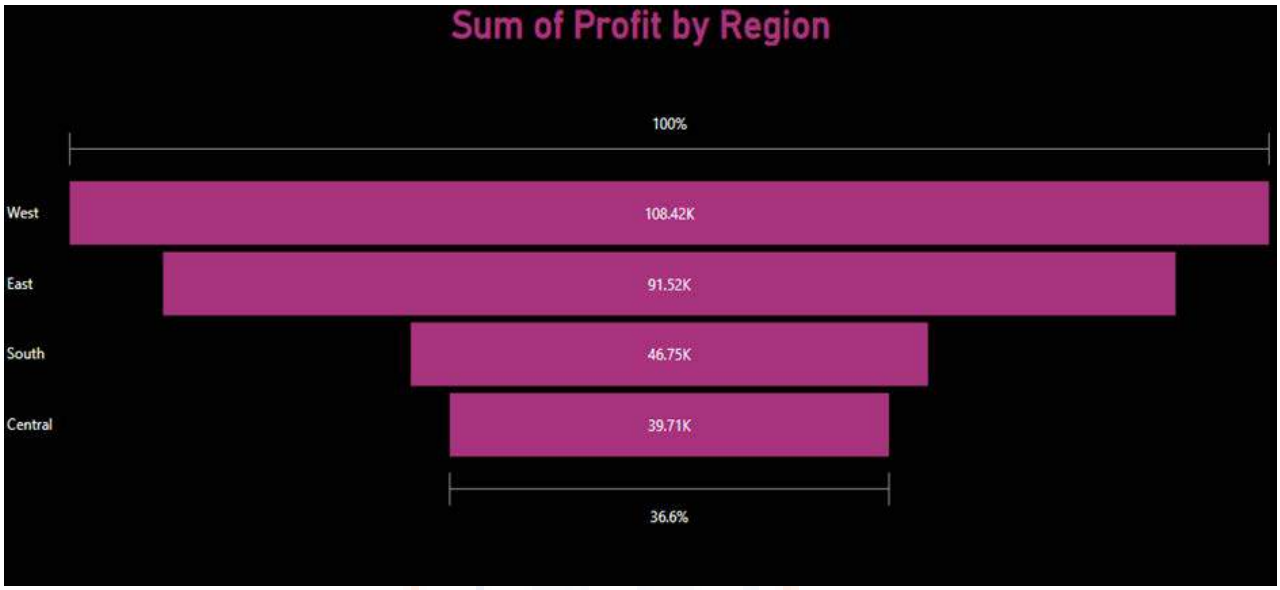
1. **Visualizing a Sequential Process**
   - Use a funnel chart to visualize a sequential process where data points progress through stages, such as sales leads moving through a sales pipeline.
2. **Analyzing Conversion Rates**
   - Ideal for analyzing conversion rates or drop-off rates at each stage of a process, helping to identify bottlenecks or areas for improvement.
3. **Highlighting Sales or Marketing Funnel**
   - Effective for highlighting the progression of sales or marketing efforts from initial leads to final conversions or sales.
4. **Comparing Performance Across Stages**
   - Useful for comparing performance metrics (e.g., conversion rates, sales volumes) across different stages of a process.

Things to Keep in Mind

1. **Data Structure**
   - Ensure your data is structured with sequential stages or steps that represent the progression of the funnel.
   - Verify that each stage has a numerical value representing the quantity or conversion rate.
2. **Order of Stages**
   - Arrange the stages in the correct order of progression, typically from the initial stage at the top to the final stage at the bottom.
3. **Color Coding**
   - Use distinct colors for each stage to differentiate them easily.
   - Maintain a consistent color scheme throughout the funnel chart.
4. **Labeling**
   - Include labels for each stage to provide context and clarify what each stage represents.
   - Ensure labels are positioned clearly and do not overlap.

5. **Width and Size**
    - Adjust the width or size of each stage to represent the quantity or value it corresponds to.
    - Ensure that the width or size accurately reflects the data and does not mislead viewers.
6. **Title and Description**
    - Provide a clear and descriptive title.
    - Include additional descriptions if necessary to clarify what each stage of the funnel represents.
7. **Interactivity and Tooltips**
    - Utilize Power BI's interactive features to allow users to drill down into more detailed data.
    - Enable tooltips to provide additional context or information when users hover over each stage of the funnel.
8. **Comparison with Other Charts**
    - Evaluate if a funnel chart is the best option or if another type (e.g., bar chart, line chart) would be more effective for your data visualization.
    - Consider how well the funnel chart communicates the progression and conversion rates across stages.
9. **Handling Incomplete Data**
    - Address incomplete or missing data appropriately, such as by excluding stages with no data or providing explanations in the chart.
10. **Data Accuracy**
    - Ensure that the data accurately reflects the progression and conversion rates at each stage of the funnel.
    - Double-check calculations and data sources to ensure accuracy.
11. **Avoiding Redundancy**
    - Avoid using a funnel chart if the data does not follow a sequential process or if the visualization does not add value to the analysis.
    - Ensure that the funnel chart effectively communicates the progression and conversion rates in a clear and intuitive manner.

Brainalyst © 2021-2024


Sum of Profit by Region
100%
West
108.42K
East
91.52K
South
46.75K
Central
39.71K
36.6%

TABLE

When to Use a Table in Power BI

1. **Detailed Data Presentation:**
   - Use a table when you need to present detailed data with multiple fields and records.
2. **Comparing Individual Records:**
   - Ideal for comparing individual records across different dimensions or attributes.
3. **Listing Raw Data:**
   - Use tables to display raw data or transactional data without summarization.
4. **Precise Data Viewing:**
   - Effective for users who need to view precise values and drill down into specific records.
5. **Data Exploration:**
   - Use tables for data exploration and analysis, allowing users to interactively filter, sort, and search for specific data points.

Things to Keep in Mind

1. **Column Selection:**
   - Select relevant columns to include in the table based on the analysis requirements.
   - Avoid including too many columns to prevent clutter and improve readability.
2. **Data Sorting:**
   - Enable sorting options to allow users to sort the data based on different columns.
   - Consider setting default sorting to present data in a meaningful way.
3. **Column Formatting:**
   - Format columns appropriately to ensure data is displayed correctly (e.g., numeric values with appropriate decimal places, date formats).
   - Use conditional formatting to highlight important data or trends.
4. **Row Limitation:**
   - Limit the number of rows displayed in the table, especially if dealing with large datasets.
   - Implement pagination or scroll bars to improve performance and usability.

Brainalyst © 2021-2024

5. **Headers and Totals:**
    - Include column headers and row totals if necessary for better context and understanding of the data.
    - Ensure headers are descriptive and aligned with the data they represent.
6. **Interactivity:**
    - Make the table interactive by enabling features like filtering, sorting, and highlighting.
    - Utilize drill-through functionality to provide more detailed information on specific data points.
7. **Visual Hierarchy:**
    - Use formatting options to create visual hierarchy within the table, emphasizing important data or columns.
    - Highlight key metrics or outliers to draw attention to critical information.
8. **Data Refresh:**
    - Ensure that the data in the table is regularly refreshed to reflect the latest information from the underlying data source.
9. **Accessibility:**
    - Design the table with accessibility in mind, ensuring that it is readable and navigable for all users.
    - Consider users with color vision deficiencies and provide alternative ways to differentiate data points.
10. **Testing and Validation:**
    - Test the table with sample datasets to ensure that it meets the analysis requirements and presents data accurately.
    - Validate data calculations and aggregations to avoid errors or discrepancies in the table.

Region	Sum of Profit
West	1,08,418.45
East	91,522.78
South	46,749.43
Central	39,706.36
Total	2,86,397.02

MATRIX

When to Use a Matrix in Power BI

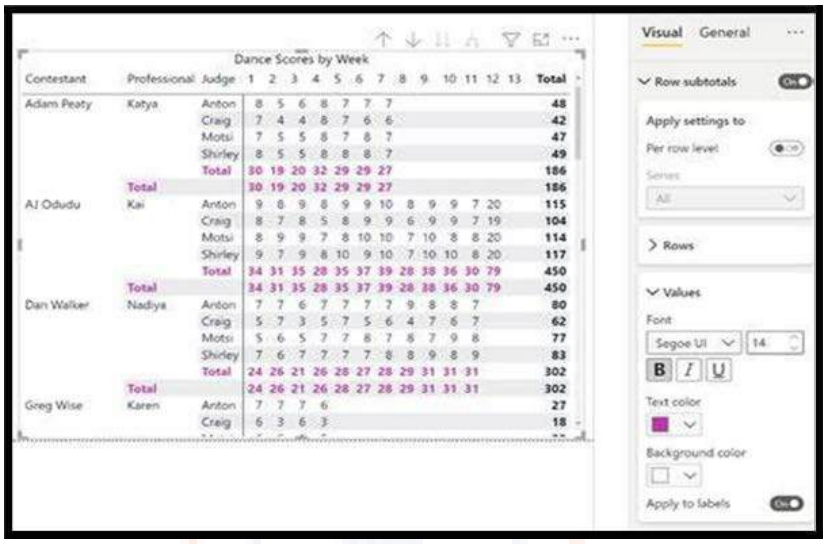
1. **Cross-Dimensional Analysis:**
   - Use a matrix to perform cross-dimensional analysis by displaying data summarized across two or more dimensions, such as categories and time periods.
2. **Comparing Multiple Metrics:**
   - Ideal for comparing multiple metrics or measures across different dimensions, allowing for a comprehensive analysis of data.
3. **Drilling Down into Data:**
   - Effective for drilling down into hierarchical data structures, enabling users to explore data at different levels of detail.
4. **Pivot Table Functionality:**
   - Offers similar functionality to pivot tables in Excel, making it suitable for users familiar with that interface.

Things to Keep in Mind

1. **Row and Column Selection:**
   - Select appropriate rows and columns to include in the matrix based on the analysis requirements.
   - Consider hierarchies and groupings to organize data effectively.
2. **Data Aggregation:**
   - Define appropriate aggregation functions for each data field to summarize data accurately (e.g., sum, average, count).
3. **Subtotals and Grand Totals:**
   - Include subtotals and grand totals if necessary to provide a summary of data at different levels of aggregation.
   - Ensure subtotals are calculated correctly based on the hierarchy of the data.
4. **Formatting Options:**
   - Apply formatting options to improve the visual appearance of the matrix, such as font styles, colors, and borders.
   - Use conditional formatting to highlight important data points or trends.

Brainalyst © 2021-2024

5. **Interactivity:**
    - Make the matrix interactive by enabling features like sorting, filtering, and expanding/ collapsing rows and columns.
    - Utilize drill-through functionality to provide more detailed information on specific data points.
6. **Hierarchy Management:**
    - Manage hierarchies effectively by collapsing or expanding rows and columns to focus on relevant levels of detail.
    - Implement sorting options to allow users to arrange data hierarchically or by specific criteria.
7. **Subtotals and Grand Totals:**
    - Include subtotals and grand totals if necessary to provide a summary of data at different levels of aggregation.
    - Ensure subtotals are calculated correctly based on the hierarchy of the data.
8. **Data Labels and Tooltips:**
    - Include data labels or tooltips to provide additional context and information when users hover over data cells.
    - Ensure labels are clear and concise to aid understanding.
9. **Performance Optimization:**
    - Optimize the performance of the matrix by limiting the number of rows and columns displayed, especially with large datasets.
    - Consider using filters and slicers to focus on specific subsets of data and improve responsiveness.
10. **Testing and Validation:**
    - Test the matrix with sample datasets to ensure that it meets the analysis requirements and presents data accurately.
    - Validate data calculations and aggregations to avoid errors or discrepancies in the matrix.



Brainalyst © 2021-2024

CARD

When to Use a Card in Power BI

1. **Single Metric Presentation:**
   - Use a card to present a single key metric or KPI prominently, providing a quick snapshot of performance or status
2. **Highlighting Important Information:**
   - Ideal for highlighting important information or key figures that require immediate attention
3. **Comparing Metrics Across Categories:**
   - Effective for comparing metrics or measures across different categories or dimensions, especially when used in conjunction with slicers or filters.
4. **Dashboard Summary:**
   - Use cards to summarize essential information on dashboards, allowing users to quickly grasp key insights without extensive analysis.

Things to Keep in Mind

1. **Metric Selection:**
   - Select the most relevant metric or KPI to display on the card, ensuring it aligns with the analysis objectives and user needs.
2. **Formatting Options:**
   - Apply formatting options to the card to improve its visual appearance and readability, such as font size, color, and style.
   - Use conditional formatting to dynamically change the appearance of the card based on specific conditions or thresholds.
3. **Precision vs. Clarity:**
   - Balance precision with clarity when formatting the card. Ensure that the displayed value is precise enough to convey meaningful information but not overly detailed to overwhelm users.
4. **Contextual Information:**
   - Provide contextual information or additional context alongside the metric displayed on the card, such as a trend indicator, comparison to a target or benchmark, or a brief description of the metric.

5. **Interactivity:**
    - Make the card interactive by enabling features like drill-through functionality or linking it to other visuals or reports.
    - Utilize tooltips to provide additional information or insights when users hover over the card.
6. **Dynamic Data Updates:**
    - Ensure that the data displayed on the card is dynamically updated as new data becomes available. Schedule regular data refreshes to keep the information current.
7. **Placement and Layout:**
    - Consider the placement and layout of the card within the report or dashboard to optimize visibility and accessibility.
    - Group related cards together to provide a cohesive view of related metrics or KPIs.
8. **Accessibility:**
    - Design the card with accessibility in mind, ensuring that it is readable and navigable for all users, including those with visual impairments.
    - Provide alternative text or descriptions for users who rely on screen readers.
9. **Testing and Validation:**
    - Test the card with sample datasets to ensure that it accurately represents the metric or KPI and aligns with user expectations.
    - Validate data calculations and formatting to avoid errors or discrepancies in the card.

Brainalyst © 2021-2024

SLICER

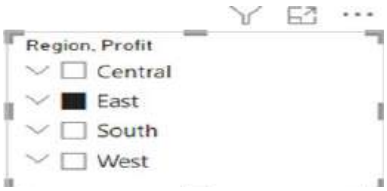
When to Use a Slicer in Power BI

1. **Data Filtering:**
   - Use a slicer to filter data interactively based on specific criteria, allowing users to explore different subsets of data.
2. **Dashboard Interactivity:**
   - Ideal for adding interactivity to dashboards, enabling users to dynamically adjust data views and drill down into specific details.
3. **Multi-Field Selection:**
   - Effective for selecting multiple values across different fields or dimensions simultaneously, providing flexibility in data analysis.
4. **Comparing Scenarios:**
   - Use slicers to compare different scenarios or segments of data, facilitating comparative analysis and decision-making.

Things to Keep in Mind

1. **Slicer Type:**
   - Choose the appropriate slicer type based on the nature of the data:
     - List Slicer: Suitable for selecting individual values from a list.
     - Dropdown Slicer: Ideal for conserving space on the report canvas while still allowing users to select values.
     - Range Slicer: Use for selecting numeric or date ranges, such as sales amounts or date ranges.
2. **Field Selection:**
   - Select relevant fields or dimensions to include in the slicer based on the analysis requirements.
   - Avoid including too many fields to prevent clutter and improve usability.
3. **Formatting Options:**
   - Apply formatting options to the slicer to enhance its visual appearance and align with the report's overall design aesthetic.
   - Customize the font size, color, style, and layout to improve readability and user experience.

4. **Single vs. Multi-Select:**
    - Determine whether users should be allowed to select a single value or multiple values from the slicer.
    - Consider the analysis context and user preferences when making this decision.
5. **Filtering Behavior:**
    - Configure the filtering behavior of the slicer to determine how selected values affect other visuals on the report canvas.
    - Choose between basic filtering (e.g., include or exclude selected values) and advanced filtering options (e.g., cross-filtering, hierarchies).
6. **Visual Hierarchy:**
    - Arrange slicers logically and hierarchically to facilitate intuitive navigation and selection of values.
    - Group related slicers together or organize them into categories for easier access.
7. **Clear Selection Option:**
    - Provide an option to clear selections or reset the slicer to its default state, allowing users to start fresh or remove applied filters.
8. **Interactivity:**
    - Make slicers interactive by enabling features like search, multi-level filtering, and dependent slicer functionality.
    - Utilize tooltips to provide additional information or guidance on slicer usage.
9. **Performance Optimization:**
    - Optimize slicer performance by limiting the number of distinct values or categories displayed, especially with large datasets.
    - Implement server-side filtering to improve responsiveness and reduce client-side processing.
10. **Accessibility:**
    - Design the slicer with accessibility in mind, ensuring that it is navigable and usable for all users, including those with disabilities.
    - Provide alternative navigation options for users who rely on screen readers or keyboard navigation.



Brainalyst © 2021-2024

GAUGE

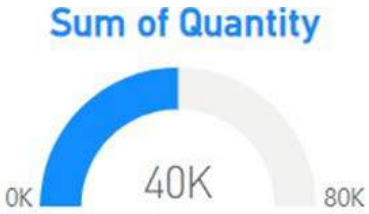
When to Use a Gauge in Power BI

1. **Visualizing Performance Against Targets:**
   - Use a gauge to visually represent performance metrics against predefined targets or thresholds, providing a clear indication of progress or attainment.
2. **Highlighting Key Metrics:**
   - Ideal for highlighting key metrics or KPIs that require immediate attention, such as sales revenue, profit margins, or customer satisfaction scores.
3. **Single Metric Presentation:**
   - Effective for presenting a single metric or value in a visually appealing and easy-to-understand format, making it suitable for executive dashboards or summary reports.
4. **Comparing Performance Across Categories:**
   - Use multiple gauges to compare performance metrics across different categories or dimensions, allowing for quick comparisons and analysis.

Things to Keep in Mind

1. **Metric Selection:**
   - Select the most relevant metric or KPI to display on the gauge, ensuring it aligns with the analysis objectives and user needs.
2. **Target Definition:**
   - Define clear targets or thresholds for the gauge to indicate desired performance levels or goals.
   - Ensure that targets are meaningful and achievable, providing context for interpreting gauge readings.
3. **Gauge Type:**
   - Choose the appropriate gauge type based on the nature of the metric and the desired visual representation:
     - **Linear Gauge:** Suitable for displaying metrics that have a clear minimum and maximum range, such as temperature or progress towards a goal.
     - **Radial Gauge:** Ideal for visualizing metrics in a circular format, such as speedometers or completion percentages.
     - **Bullet Graph:** Use for comparing actual performance against target values and ranges, providing more context than a traditional gauge.

4. **Formatting Options:**
    - Apply formatting options to the gauge to improve its visual appearance and readability, such as colors, labels, and scales.
    - Customize the gauge's appearance to match the report's overall design aesthetic and branding.
5. **Thresholds and Ranges:**
    - Define multiple thresholds or ranges on the gauge to highlight different levels of performance (e.g., red for below target, yellow for near target, green for on or above target).
    - Ensure that thresholds are clearly differentiated and provide actionable insights for users.
6. **Interactivity:**
    - Make the gauge interactive by enabling features like tooltips or drill-through functionality to provide additional context or details.
    - Utilize dynamic thresholds or targets that update based on user selections or changes in underlying data.
7. **Accessibility:**
    - Design the gauge with accessibility in mind, ensuring that it is readable and navigable for all users, including those with disabilities.
    - Provide alternative text or descriptions for users who rely on screen readers or assistive technologies.
8. **Testing and Validation:**
    - Test the gauge with sample datasets to ensure that it accurately represents the metric or KPI and aligns with user expectations.
    - Validate gauge readings against the underlying data to avoid errors or discrepancies.



LINE AND CLUSTERED COLUMN

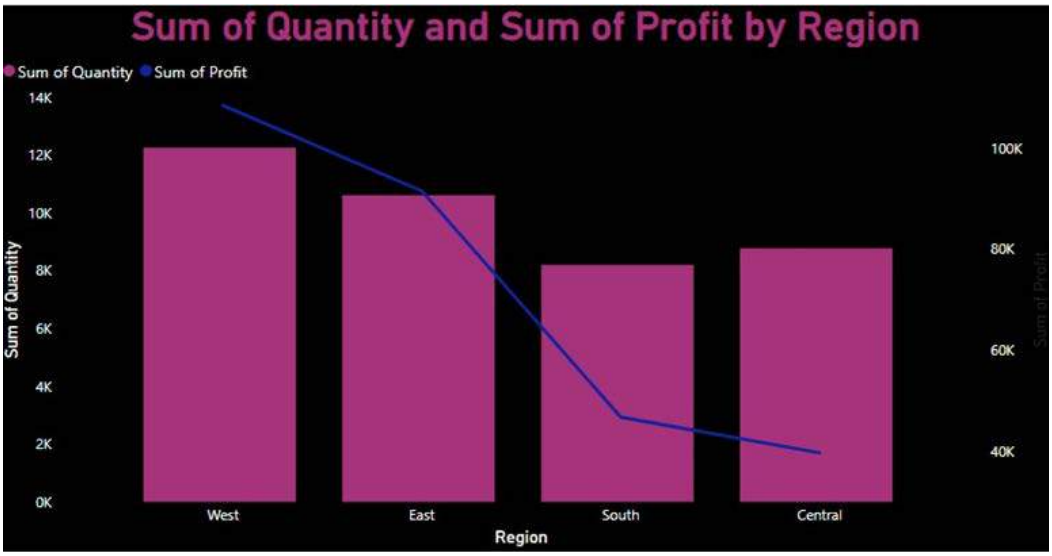
When to Use a Line and Clustered Column Chart in Power BI

1. **Comparing Trends and Totals:**
   - Use a combination of line and clustered column chart when you need to compare both trends over time and total values across different categories simultaneously.
2. **Visualizing Multiple Metrics:**
   - Ideal for visualizing multiple metrics or measures where one axis (e.g., the y-axis) represents continuous data (e.g., sales revenue) and the other axis (e.g., the x-axis) represents categorical data (e.g., time periods or product categories).
3. **Highlighting Relationships:**
   - Effective for highlighting relationships or correlations between two different types of data (e.g., sales volume and profit margin) within the same chart.
4. **Drilling Down into Data:**
   - Use drill-down functionality to enable users to explore data at different levels of detail, such as by year, quarter, month, or product category.

Things to Keep in Mind

1. **Data Structure:**
   - Ensure that your data is structured appropriately with one axis representing categorical data (e.g., time periods or categories) and the other axis representing continuous data (e.g., numerical values).
2. **Axis Configuration:**
   - Label the x-axis with categorical data (e.g., time periods) and the y-axis with continuous data (e.g., numerical values).
   - Ensure that the axes are correctly scaled to accurately represent the data.
3. **Color Coding:**
   - Use distinct colors for the line and clustered column series to differentiate them easily.
   - Maintain a consistent color scheme across similar charts for better comparison.
4. **Data Labels:**
   - Include data labels or tooltips to provide precise values for important points on the chart, such as the exact values of data points or totals.

5. **Legend and Guide:**
    - Include a legend to explain the colors and styles of the line and clustered column series.
    - Position the legend in a way that does not clutter the chart.
6. **Interactivity and Tooltips:**
    - Utilize Power BI's interactive features to allow users to filter and drill down into more detailed data.
    - Enable tooltips to provide additional context or information when users hover over data points.
7. **Title and Description:**
    - Provide a clear and descriptive title that summarizes the chart's purpose and the data it represents.
    - Include additional descriptions if necessary to clarify what each axis and series represent.
8. **Handling Complexity:**
    - Be cautious when dealing with a large number of data series or categories, as the chart can become cluttered and difficult to interpret.
    - Consider using filters or aggregating data to manage complexity and maintain readability.
9. **Comparison with Other Charts:**
    - Evaluate if a combination of line and clustered column chart is the best option or if another type (e.g., stacked column chart, area chart) would be more effective for your data analysis.
    - Consider how well the chart communicates trends, patterns, and relationships between different types of data.



Brainalyst © 2021-2024

Waterfall Chart

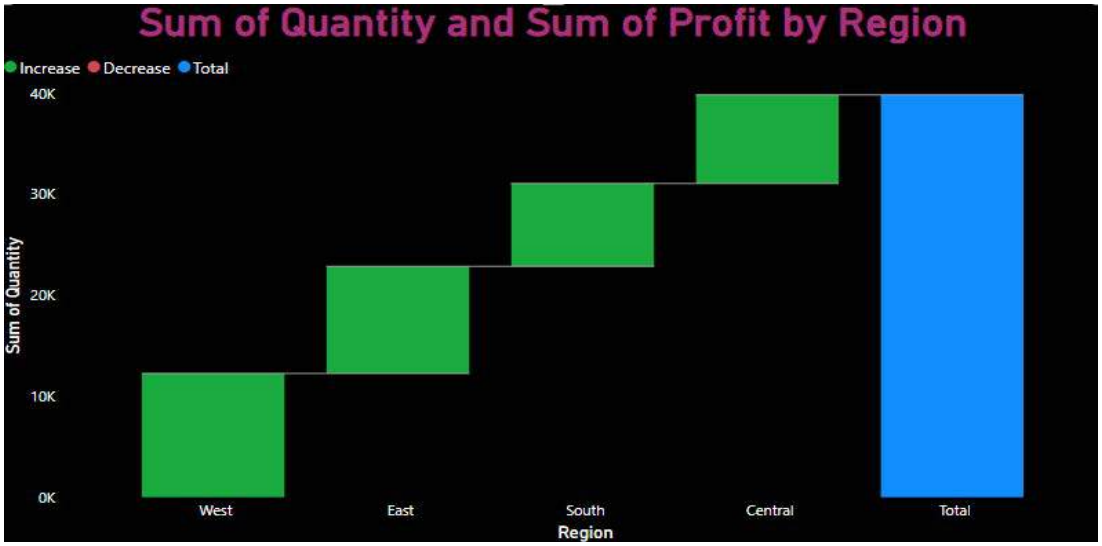
When to Use a Waterfall Chart in Power BI

1. **Analyzing Changes Over Time:**
   - Use a waterfall chart to visualize changes in values over time or across different categories, highlighting the cumulative effect of positive and negative contributions.
2. **Showing Flow of Data:**
   - Ideal for illustrating the flow or progression of data from one state to another, such as changes in revenue from sales, expenses, and net profit.
3. **Identifying Contributors to Total:**
   - Effective for identifying the individual contributors to a total value, breaking down the components that make up the overall change.
4. **Highlighting Trends and Variances:**
   - Use a waterfall chart to highlight trends, variances, and the magnitude of changes between different data points.

Things to Keep in Mind

1. **Data Structure:**
   - Ensure that your data is structured appropriately with clear categories or stages representing the flow of values (e.g., initial value, positive and negative changes, final total).
2. **Start and End Points:**
   - Define the starting and ending points of the waterfall chart, indicating the initial value and the final total.
   - Ensure that the starting point is clearly labeled and represents the baseline or reference point for the chart.
3. **Positive and Negative Changes:**
   - Differentiate between positive and negative changes by using different colors or formatting options (e.g., green for increases, red for decreases).
4. **Intermediate Values:**
   - Include intermediate values or segments to represent individual contributions to the total change.
   - Label each segment to provide context and clarity on the nature of the change (e.g., sales increase, expenses decrease).

5. **Totals and Subtotals:**
    - Include totals and subtotals to summarize the cumulative effect of changes and provide context for interpreting the overall impact.
6. **Axis Scaling:**
    - Adjust the axis scaling to ensure that all segments of the waterfall chart are visible and accurately represented.
    - Consider using logarithmic scaling if there are significant variations in the magnitude of changes.
7. **Formatting Options:**
    - Apply formatting options to the waterfall chart to improve its visual appearance and readability, such as colors, labels, and borders.
    - Customize the chart's appearance to match the report's overall design aesthetic and branding.
8. **Interactivity:**
    - Make the waterfall chart interactive by enabling features like tooltips or drill-through functionality to provide additional context or details.
    - Utilize dynamic filtering to allow users to focus on specific categories or time periods.
9. **Title and Description:**
    - Provide a clear and descriptive title that summarizes the chart's purpose and the data it represents.
    - Include additional descriptions if necessary to explain the methodology or assumptions used in calculating the changes.



Brainalyst © 2021-2024

DECOMPOSITION TREE

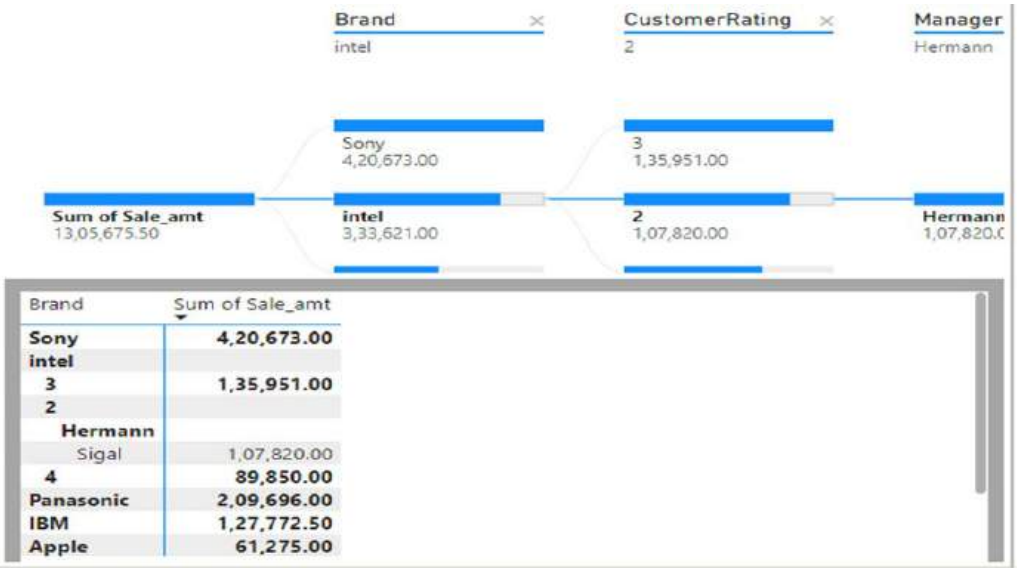
When to Use a Decomposition Tree in Power BI

1. **Exploring Hierarchical Data:**
   - Use a decomposition tree to explore hierarchical data structures with multiple levels of aggregation or categorization.
2. **Understanding Data Breakdown:**
   - Ideal for understanding the breakdown of a metric or measure across different dimensions, such as time periods, product categories, or geographic regions.
3. **Identifying Key Contributors:**
   - Effective for identifying key contributors to a metric or measure and understanding their relative importance or impact.
4. **Root Cause Analysis:**
   - Use the decomposition tree for root cause analysis to drill down into the underlying factors driving changes or variations in a metric.

Things to Keep in Mind

1. **Data Structure:**
   - Ensure that your data is structured hierarchically with clear relationships between different levels of aggregation or categorization.
   - Verify that the data contains relevant dimensions or attributes for decomposition.
2. **Root Level Selection:**
   - Choose the appropriate root level for the decomposition tree, indicating the starting point for analyzing the data breakdown.
   - Select a root level that provides meaningful insights into the metric or measure being analyzed.
3. **Metric Selection:**
   - Select the metric or measure that you want to analyze and decompose using the tree.
   - Ensure that the metric is relevant and meaningful for the analysis objectives.
4. **Dimension Selection:**
   - Choose the dimensions or attributes that you want to use for decomposing the selected metric.
   - Select dimensions that provide useful insights into the factors contributing to changes in the metric.

5. **Drill Down and Drill Up:**
    - Use drill-down and drill-up functionality to navigate through different levels of aggregation or categorization within the decomposition tree.
    - Drill down to explore detailed breakdowns of the selected metric, and drill up to view higher-level summaries.
6. **Sorting and Filtering:**
    - Apply sorting and filtering options to focus on specific categories or segments within the decomposition tree.
    - Sort categories based on relevant criteria such as contribution to the selected metric or alphabetical order.
7. **Visualization Options:**
    - Customize the visualization options for the decomposition tree, such as colors, labels, and formatting.
    - Adjust the appearance of the tree to improve its visual clarity and readability.
8. **Interactivity:**
    - Make the decomposition tree interactive by enabling features like tooltips, selection, and highlighting.
    - Utilize dynamic filtering to allow users to focus on specific categories or segments of interest.
9. **Title and Description:**
    - Provide a clear and descriptive title for the decomposition tree, indicating the metric or measure being analyzed and the dimensions used for decomposition.
    - Include additional descriptions or annotations to provide context and guidance for users.



Brainalyst © 2021-2024